

Business Brief

Sales Distribution Analysis for a Retail Business

1. Business Background

The company is a retail business that sells products to customers across different cities, states, and regions. The business uses different shipping methods and serves different customer segments.

Management has collected transaction-level sales data containing information about orders, shipping, customer segments, locations, product categories, and sales values.

The dataset contains the following key fields:

- **Order ID** – unique identifier for each order
- **Ship Date** – date the order was shipped
- **Ship Mode** – method used to ship the order
- **Segment** – customer segment
- **City** – customer city
- **State** – customer state
- **Region** – geographical region
- **Category** – broad product category
- **Sub-category** – specific product group
- **Sales** – value of the sale

The dataset is prepared for **descriptive analysis**, with the aim of understanding what has happened in the company's sales performance.

2. Problem Statement

Management has sales information but does not have a simple summary showing how sales are distributed across products, customers, locations, and shipping methods.

Although individual transactions are available, it is difficult to quickly determine:

- The total sales generated
- Which categories generate the most sales
- Which regions contribute the most sales

- How sales are distributed across customer segments
- Which shipping modes account for the largest share of sales
- How sales are distributed across different orders

Without a clear summary, management may find it difficult to communicate the company's current sales performance effectively.

3. Objectives of the Analysis

The objective of this analysis is to use descriptive analytics to summarize **WHAT is happening** in the sales data.

Specifically, the analysis will:

1. Determine the **total sales value** recorded in the dataset.
 2. Determine the **number of unique orders**
 3. Show how sales are distributed across **Category**
 4. Analyze sales distribution across **Region**
 5. Examine sales across different **Customer Segments**.
 6. Analyze sales by **Ship Mode**.
 7. Present the findings using simple tables, charts, and dashboard.
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4. Primary KPIs

To keep the analysis simple and focused, the dashboard should emphasize two primary KPIs:

KPI 1: Total Sales

Total Sales = Sum of Sales

This shows the total monetary value of all sales recorded in the dataset.

KPI 2: Number of Orders

Number of Orders = Count of Unique Order IDs

This shows how many individual orders are represented in the dataset.

These two KPIs should be displayed prominently at the top of the dashboard.

5. Key Business Questions

A. Sales Overview

- What is the total sales value recorded in the dataset?
- How many unique orders are recorded?
- What is the average sales value per order?

B. Category & Product Distribution

- What is the total sales generated by each **Category**?
- Which Category contributes the largest share of total sales?

C. Geographic Distribution

- What is the total sales generated by each **Region**?
- Which Region contributes the largest share of sales?

D. Customer Segment Distribution

- What is the total sales generated by each **Segment**?
- Which customer Segment contributes the largest share of sales?

E. Shipping Distribution

- What is the total sales generated by each **Ship Mode**?
- Which Ship Mode accounts for the largest share of total sales?

F. Time/Shipping-Date View

- How are sales distributed by **Ship Date**?
- Which months or periods recorded the highest sales?

6. Expected Deliverables

The completed analysis should contain:

- A **Sales Overview** showing Total Sales and Number of Orders
- Pivot tables for the major business dimensions
- A **Sales by Category** chart
- A **Sales by Region** chart
- A **Sales by Segment** chart

- A **Sales by Ship Mode** chart
- A **Sales by Ship Date** trend
- A simple interactive dashboard